Patch-level Phenotype Identification via Weakly Supervised Single-neuron Selection in Sparse Autoencoders for CLIP-derived Pathology Embeddings

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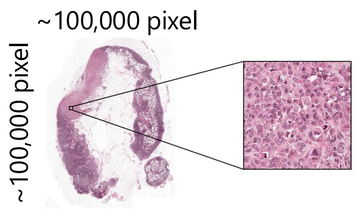
Introduction

Challenge in WSI Analysis

- Whole-slide images (WSIs) are gigapixel-scale pathology scans with only slide-level labels, making patch-level supervision infeasible.



Limitations of Current MIL Approaches

- **Feature Entanglement** – High-dimensional features are difficult to interpret biologically.
- **Slide-relative Attention** – Attention weights are normalized within each slide, hindering cross-slide comparability and limiting consistent patch-level phenotype identification.

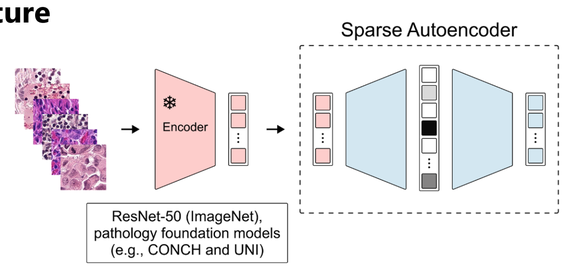
Proposal

- We propose SAE-1N, a weakly supervised framework that employs a Sparse Autoencoder (SAE) to disentangle CLIP-derived pathology embeddings and identify single neurons associated with distinct histopathological phenotypes at the patch level.

Methods

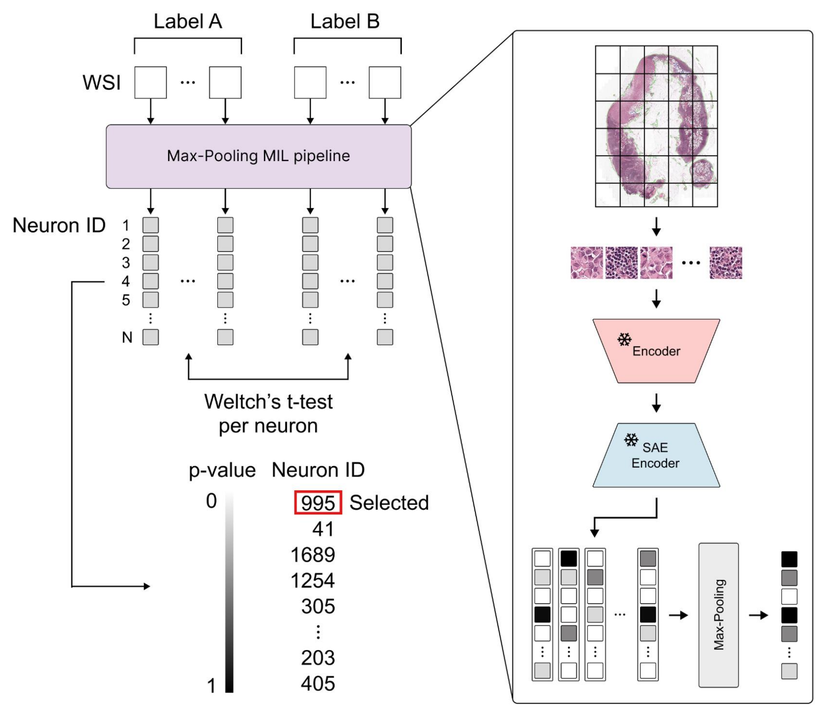
Sparse Autoencoder (SAE) Architecture

- Projects CLIP-derived pathology embeddings into an overcomplete, sparse latent space.
- Promotes disentangled, interpretable neuron activations reflecting distinct morphological patterns.



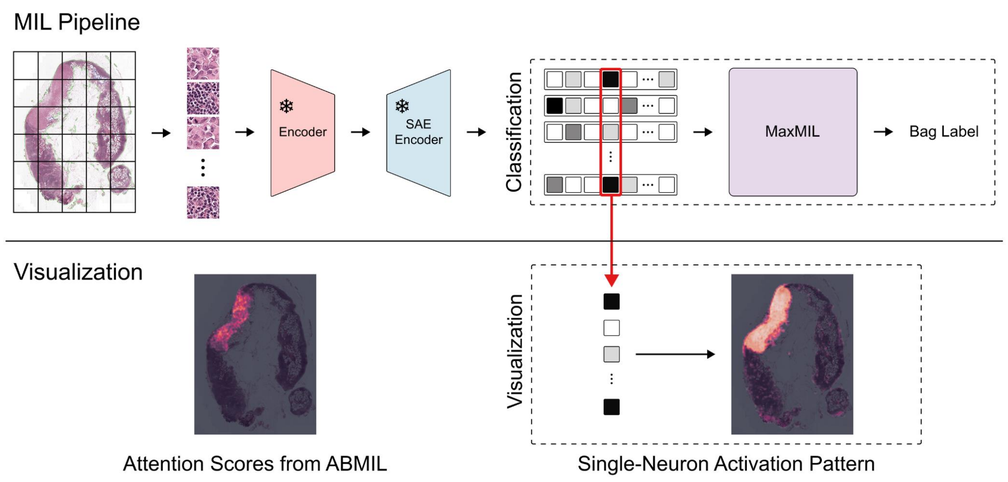
Weakly Supervised Neuron Selection (SAE-1N)

- Aggregates patch activations per slide and applies statistical testing (Welch's t-test) to detect phenotype-associated neurons.
- Selects the top-1 significant neuron as a single interpretable signal.



Phenotype Identification

- Visualizes the activation map of the selected neuron to highlight patch-level phenotype regions.
- Enables direct, interpretable linkage between neuron activity and histopathological features.



Results

Datasets

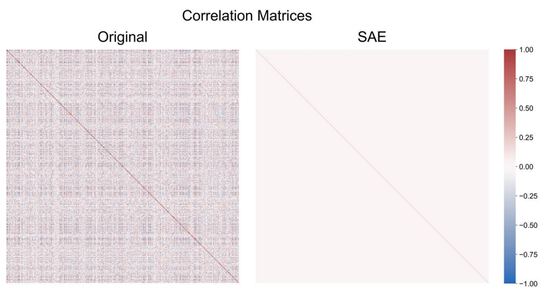
- **Camelyon16** – 399 WSIs of lymph node metastases for binary tumor detection.
- **PANDA** – more than 10,000 prostate WSIs, reformulated as a binary tumor vs. normal task for consistency.

Encoder: CONCH

- CLIP-style vision-language model trained on 1.17M pathology image–caption pairs.
- Produces 512-dimensional, semantically enriched embeddings aligned with textual pathology descriptions.

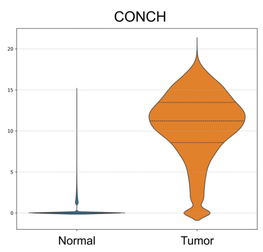
Feature Disentanglement

- SAE transformation reduces inter-neuron correlation, indicating more disentangled representations.



Patch-level Phenotype Identification

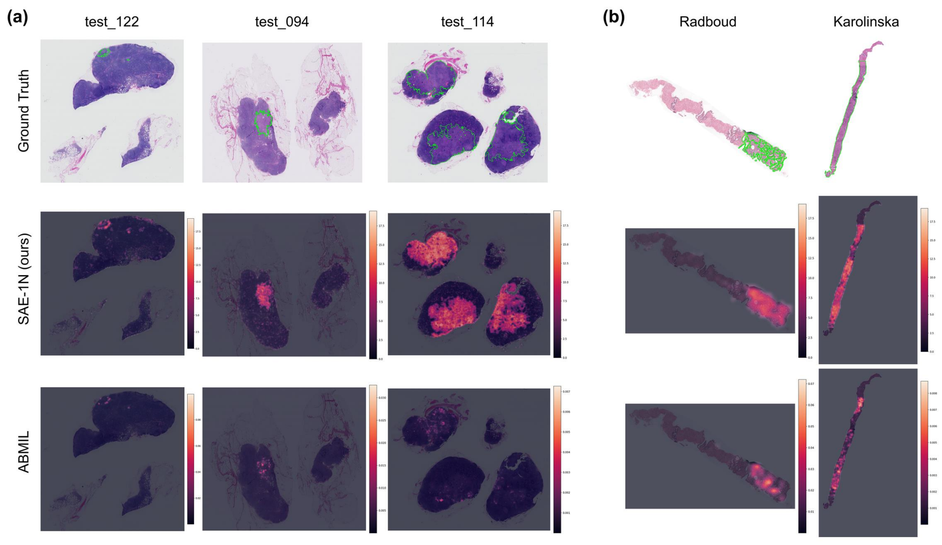
- Single-neuron activations identified by SAE-1N clearly separate Tumor vs. Normal patches on Camelyon16.
- The selected neuron yields accurate and interpretable localization of tumor regions, outperforming attention-based MIL models.



Method	Camelyon16	PANDA		
		R&K	Radboud	Karolinska
Zero-shot (CONCH)	0.1612 (± 0.0107)	0.5361	0.5969	0.4990
ABMIL	0.5146 (± 0.0482)	0.5606	0.6198	0.5017
CLAM-SB	0.4785 (± 0.0630)	0.5563	0.5822	0.4767
CLAM-MB	0.5551 (± 0.0208)	0.5781	0.5596	0.4945
TransMIL	0.0950 (± 0.0269)	0.4566	0.4443	0.4549
SAE-1N	0.6135 (± 0.0309)	0.5705	0.6375	0.5129

Interpretability

- SAE-1N provides more comprehensive visualization and uses an absolute activation scale, addressing key limitations of attention-based methods.



Conclusion

- SAE-1N effectively disentangles CLIP-derived pathology embeddings, enabling transparent, patch-level phenotype identification through single-neuron activations.
- The approach achieves competitive accuracy while offering standardized and interpretable visualization based on absolute activation scales.
- Future Work – Extend the neuron selection strategy to multi-phenotype settings and explore broader diagnostic generalization.

Acknowledgements

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